

☆ Electric Vehicle

defn: uses one or more electric motor for propulsion

→ does not rely completely on ICE

→ Mechanical energy for vehicle movement
is produced by electric motors

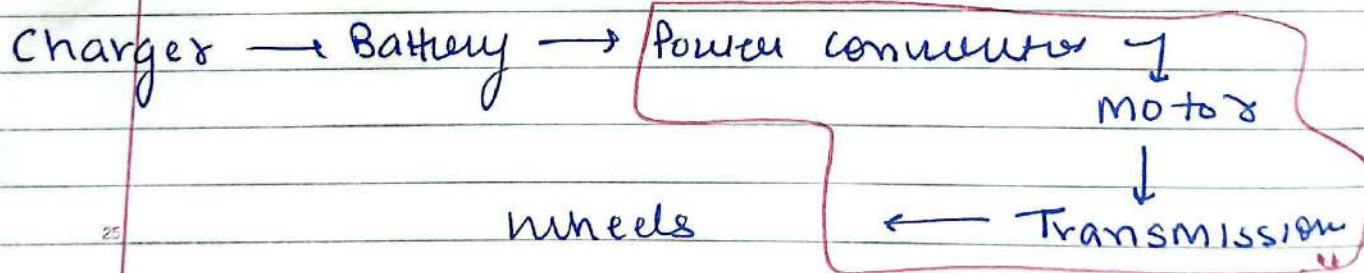
→ Electricity is supplied by battery packs

Electric energy $\longleftrightarrow$ mechanical energy
↓
through electric motors

High η & zero tailpipe emissions

★ Construction:

BASIC Power flow



① Battery:

- # Li-Ion cells (usually)
- # High energy density
- # long life cycle
- # Supplies DC electricity

2) # Battery capacity determines driving range

② Electric Motor :

Replaces IC engine

electricity $\rightarrow$ magnetic field $\rightarrow$ causes rotor to rotate

Rotational motion is transferred to wheels

can use AC or DC motors

AC motors } often combined with inverters for speed & torque control

③ Electronic Controller

Controls & monitors EV operation

Regulates electrical energy from battery to motor

Responds to accelerator i/p

Controls motor speed & torque

Also works with regenerative braking

④ BRAKING SYSTEM

Uses Regenerative Braking

Motor acts as generator ~~using~~ during braking

Mechanical energy $\xrightarrow{\text{converted}}$ electrical energy

Recovered energy is stored in battery

Improves range & efficiency

⑤ BATTERY CHARGER

- # Recharges the battery from an external supply.
- # Acts as a rectifier & voltage controlled device.

AC → DC

- # Provides controlled V_g for battery charging

★ Advantages Vs Disadvantages

Advantages	Disadvantages
① Environmental Benefits	① High Upfront Cost
② Lower Running & maintenance cost	② Range Anxiety
③ Performance & Driving Experience.	③ Charging & Infra Limitation
④ Convenience	④ Battery Issues
⑤ Energy Security & Sustainability	⑤ Performance Limitations
⑥ Incentives & Technology Advancements	⑥ Limited Vehicle Choice
	⑦ Environmental Impact of Manufacturing

★ ICE

① WORKING:

chemical energy (petrol/diesel) $\xrightarrow{\text{internal combustion}}$

Thermal energy.

Thermal energy produces piston movement

piston drives the crankshaft

crankshaft connects to mechanical T-X

T-X controls torque delivered to differential & wheels

② Efficiency & energy loss

well to wheel

thermal efficiency: 20-35%

more than 65% energy is lost

losses occur through

- Exhaust heat
- Cooling systems
- Mechanical friction

③ Emissions

Products $\xleftarrow{\text{CO}_2}$ $\xleftarrow{\text{NO}_x}$ $\xleftarrow{\text{CO}}$
unburn hydrocarbons

★ Types of EV

① HEV (Hybrid Electric Vehicle)

define: # Combines IC + Electric motor + small high voltage traction battery.

typical Battery: 1-2 kWhr.

cannot be plugged into the grid

Battery is charged by

- Regenerative Braking
- Engine-driven generator

Uses both fossil fuel + electricity.

ICE + motor work together to optimize efficiency & performance

Working Conditions:

Low speed Driving

- Electric motor propels vehicle
- Battery supplies motor
- low acceleration requirement
- IC machine remains disengaged.

High speed / Heavy Acceleration

- when climbing hill or highways
- Both ICE / motor operate
- ICE] main power
- Motor supplements ICE

- # Braking:
- motor act as a generator
 - converts vehicle motion into electricity
 - Electricity is stored in Battery.

- Features:
- # Better Performance & fuel efficiency
 - # ICE does not operate all time
 - # Reduces greenhouse gas emissions & pollution
 - # long drive range.

Efficiency : $\sim 35-45\%$ (thermal efficiency).

② PHEV (Plug In Hybrid EV).

- defn:
- # advanced HEV
 - # Construction is similar to HEV
 - # Has an external charging system
 - # Battery can be charged

Operation Modes:

All Electric Mode

HYBRID Mode

When battery is sufficiently discharged, controller switches to hybrid mode.

Features: # No noise

Hybrid mode provides heavy acceleration & high speeds

longer driving range

Regenerative braking recharges battery.

Higher efficiency & lower fuel consumption.

③ BEV (Battery Electric Vehicle)

define # Uses only electric motors powered by rechargeable batteries

No petrol

complete dependent on electricity

Main Components

Battery pack

Onboard charger

Inverter

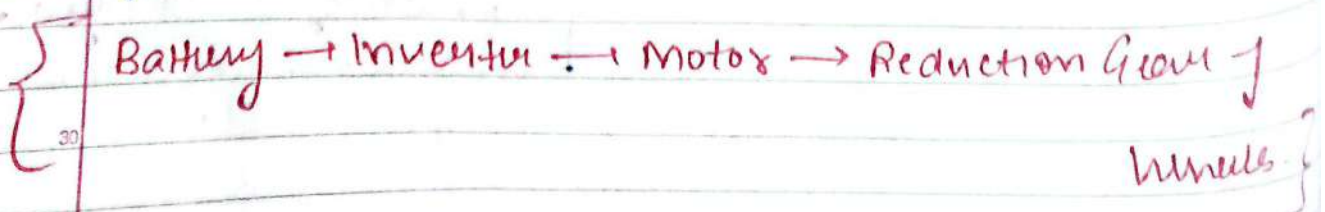
Electric Motor

Reduction gear

Differential / axle

Vehicle control unit.

Working:



features:

- # No combustion of fossil fuel
- # Lower cost
- # Lower maintenance
- # Quiet operation
- # Suitable for many applications.

Efficiency:

(75-90%)

Minimal thermal
losses

Instant Torque from 0 RPM.

Emissions:

Zero Tailpipe emissions.

④ FCEV (fuel cell)

define # uses fuel-cell technology to generate electricity.

Electricity powers electric motors

Uses high-pressure hydrogen gas as a fuel.

Working # Hydrogen is stored in leakproof high pressure tank.

H_2 enters the fuel cell stack

O_2 comes from air

H_2 reacts with oxygen

↳ produces:

- electricity
- Heat
- water vapour.

Features # water vapour is byproduct

Onboard electricity generation

H_2 refuelling is almost as quick as petrol

High energy efficiency

Silent operation

Efficiency: 40-60%

efficiency reduces due to

- Hydrogen production
- electrolysis.

Imp

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Date : _____

efficiency :

ICE $\rightarrow$ 20-35%

HEV $\rightarrow$ 35-45%

BEV $\rightarrow$ 75-90%

FCEV $\rightarrow$ 40-60%

Energy source

ICE $\rightarrow$ Petrol

HEV $\rightarrow$ Petrol + Battery

BEV $\rightarrow$ Electricity

FCEV $\rightarrow$ Hydrogen

Tail Pipe Emissions

ICE $\rightarrow$ CO₂ + NO_x + particulates

HEV $\rightarrow$ low moderate CO₂

BEV $\rightarrow$ Zero

FCEV $\rightarrow$ Zero / water vapour

Charging / Refueling

ICE $\rightarrow$ 3-5 min

HEV $\rightarrow$ 3-5 min

BEV $\rightarrow$ 20-60 min AC / 4-8 hr DC

FCEV $\rightarrow$ 3-5 min

(L2) — Impacts

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Date :

★ Environmental : (

① Zero / Reduced Tailpipe emissions

② Climate change mitigation

③ Noise Pollr Reduction

④ Efficient energy use.

⑤ Resource conservation

⑥ Positive impact on ecosystem

⑦ Reduced dependency on fossil fuel

⑧ Lifecycle environmental impact.

Social:

① Healthier Communities

② Quicker Operation

③ Energy Independence

④ Economic Opportunities

⑤ Social Equity & Accessibility

⑥ Sustainable Urban development.

⑦ Consumer Empowerment.

⑧ Innovation Advancement

⑨ Cultural Supt

⑩ Equity Consideration

1) Economic Growth & Job Creation

2) Energy Market T+ & Investment

3) Real Estate Value Enhancement;

4) Govt Initiative & Policies

5) Supply chain Restructuring & Diversification

6) Loss of ownership & Consumer Savings

7) Grid Stability & Auxiliary Services

8) Industrial Transformation & Innovation

9) Rural development & Electrification

10) Second-life Battery Markets

11) Resilient value & Market Dynamics

Impact of EV's on Power System:

EV's obtain energy from power grid

↓
This adds a significant load.

Impact depends on

- Number of EV's
- charging power.
- charging pattern
- location of charging station
- charging duration
- Controlled or uncontrolled charging

Low EV penetration can generally be handled by the existing grid

Large scale uncontrolled charging can cause.

- Vg instability
- Harmonics
- Vg sag
- Inc power losses
- T_f overloading
- Power quality degradation.

★ Fundamental EV-Grid Dynamics

EV as a GRID load

- EV's do not generate their primary energy
- They act as grid - dependent loads
- Electricity must be generated & distributed to charge them.

load progression

more → inc demand → greater EV of electricity grid load -

Main Change

- # conventional vehicles require petrol / diesel
- # EV's replace liquid - fuel demand with electrical demand.
- # Transportation energy demand is shifted toward the power grid.

* Energy Generation & Supply Requirement :

→ Increased Generation Req. :

Large-scale EV adoption requires generation capacity

Eg : - 1 million EV's

- Each consumes 10 kWh/day
- Total = 10 million kWh/day
- = 10 GWh/day if additional demand.

→ Environmental Impact

If additional electricity comes from fossil fuels

then # CO₂ is not eliminated

emissions shift from
vehicle tailpipe → power plants,
smoke stacks.

→ Positive effects :

- ' Better integration of renewable energy
- ' Reduce oil imports
- ' Reduce greenhouse gas emissions from
clean grids.

Transmission & distribution Infrastructure

power delivery path

Power Plant $\rightarrow$ T^d line $\rightarrow$ Substations



Distribution T^d



Home / charging

stations.

major challenges:

- # Higher current-carrying capacity required
- # Inc line loading
- # Inc T^d & distribution losses
- # longer & stronger substation required.

T^d example: $\left[\begin{array}{l} 100 \text{ homes} \times 7 \text{ pu} \\ \text{charger demand} \end{array} \right] = 700 \text{ pu}$

A 500 kVA T^d can become severely overloaded.

Peak load & charging Behaviour:

NORMAL LOAD PATTERN

MORNING $\rightarrow$ MODERATE

AFTERNOON $\rightarrow$ MEDIUM

EVENING $\rightarrow$ HIGH

7PM Peak Problem

• Commuters return home.

- Many EV's start charging simultaneously
- Creates a sharp inc in evening peak load

Causes: • Vg fluctuations

- Higher generation cost
- Reduced grid reliability

Load mitigation

- Shift charging to night / off peak hrs
- Use smart charging

* Smart charging :

Intelligent control of charging time
+ charging rate

Based on : Electricity demand
Grid conditions

Electricity price

Renewable energy
availability

T/f capacity

vehicle departure time.

Eg: Day = ₹10/unit
Night = ₹5/unit

Smart charger delays charging
until ~~off-peak~~ off-peak hrs

Benefits :

- Reduces consumer cost
- Reduces grid stress
- Reduces peak demand
- Improves v/g profile
- Reduces losses

⑦ V2X Technologies

V2G $\Rightarrow$ Vehicle To Grid

- EV supplies electricity back to grid
- Helps during peak demand.

V2H $\Rightarrow$ Vehicle To Home.

- EV supplies power to house
- Useful during outages or peak demand.

V2V $\Rightarrow$ Vehicle To Vehicle

- One EV provide charge to another EV.

VPP $\rightarrow$ Virtual power plant.

- Multiple EVs operate together as one large battery
- Can provide power or energy storage to grid.

② V_g Instability

→ Ability of power system to maintain acceptable V_g during normal operation and disturbances.

Cause:

- # large scale EV charging increasing current
- # Distribution buses have impedance.
- # Increased current causes V_g drop.

$$\Delta V = IZ$$

Effect:

- # V_g fluctuations
- # V_g instability
- # V_g collapse.

MAIN CAUSES

- ① NON-LINEAR LOAD CHARACTERISTICS
- ② UNPREDICTABLE POWER CONSUMPTION
- ③ Battery Oscillations

MITIGATION

- Smart Charging
- Coordinated charging
- V_g Regulators
- Reactive - power control
- Energy storage.

Wide Area Monitoring System - (WAMS)

Purpose: # Monitors the electrical power n/w over a large geo-graphical area

Useful for (EV) demand grid for rapidly changing loads

BASIC STRUCTURE :

CT | VT $\longrightarrow$ Phasor Data Converter $\longrightarrow$ Monitoring & Control

Components :

CT : measures current

VT : measures Vg.

PMU : Provides synchronized phasor measurements

PLC : collects data from multiple PMUs

Uses :

1. Visualization

2. B/S Analysis

3. Disturbance detection

4. System Control.

5. Protection

6. Decision-making

Corrective Actions

- Reduce EV charging power
- change T/f tap settings.
- Activate energy storage
- Modify power flows

★ Harmonics & Power Quality Distortion

↓
Sinusoidal components having frequencies that are integer multiples of fundamental frequency

Eg: 50 Hz system

- 3rd → 150 Hz
- 5th → 250 Hz
- 7th → 350 Hz

Course in EVs

- EV chargers use power electronic converters
- Rectifiers & semi-conductor switches behave as non-linear loads
- They can draw non-sinusoidal current
- This produces harmonic currents & voltage distortion

★ THD : measures harmonic distortion relative to fundamental component.

$$THD_1 = \frac{\sqrt{I_1^2 + I_2^2 + \dots}}{I_1} \times 100\%$$

- High THD causes
- Increased losses
 - Heating of cables & T/F
 - Reduced insulation life
 - Protection system problems.

Reducing Harmonics

- # Power-factor Correction
- # Passive Active filters
- # Improved charger design
- # PWM converter
- # Controlled Rectifiers

★ Voltage Sag

define # Temp reduction in RMS supply Vg
Usually lasts between half cycle to one minute

Eg : 50 Hz system

• 1 cycle = 20 ms

• Half cycle = 10 ms.

Causes:

- # Sudden high demand of charging
- # Especially imp with DC fast chargers

Effects:

~~Excessive~~

- # Vg available to other consumer decreases
- # Can affect

- Sensitive electronics
- Industrial drives
- Computers
- Control systems.

Mitigation:

- # Coordinated charging
- # Vg Regulators
- # Energy Storage
- # Local renewable generation

★ EV GRID PENETRATION

★ Charger levels

20% Penetration

Level 1:

- lower Vg (110V)
- slow charging
- less Vg sag
- longer charging time

60% Penetration

Level 2:

- Higher charging power
- faster charging
- greater risk of Vg sag

★ Power losses: ($P = I^2 R$)

Effort: • EV charging increasing current

• Higher current increases losses in

↳ Feeders

↳ Cables

↳ T/f

• Loss increases with square of current

Solution: # Coordination charging

local generation

Rooftop solar PV

energy storage

★ EV Configuration

↳ Energy flows mainly through flexible electrical wires instead of rigid shafts

↳ Allows different propulsion arrangements such as independent 4 wheel drive and in wheel drive.

★ Major Subsystem:

① Electric Propulsion	② Energy Source	③ Auxiliary system
• Electronic Controller	• Battery	
• Power Converter	• Fuel cell, Hydrogenator	• Power Steering unit
• Electric Motor	• Energy Manag. unit	• Temp control unit
• Mechanical TX		• Auxiliary Power supply.
• Drivings wheels	• Energy Refuelling unit	

★ Energy & Control Flow:

Working • Brake/accelerator provides control input

- Electronic Controller generates control signals
- Power Converter regulates power flow
- Power flows thru energy source & motor backwards toward the energy source
- During regenerative braking, power flows toward the energy source
- Energy management unit controls regenerative braking & energy recovery
- It also controls refuelling & monitoring energy - source usability

- Auxiliary power supply provides required Vg levels to auxiliary system.

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A Electric Propulsion System: (Three main components)

① Electronic Motor

- converts electrical energy $\rightarrow$ mechanical energy
- Propels the vehicle
- can work in reverse during regenerative braking
- can generate electricity for charging energy storage

② Power Converter

- supplies the motor with appropriate Vg & current

③ Electronic Controller

- sends control signals to power converter
- controls motor operation
- controls required torque & speed according to driver command

A Electronic Controller

① Sensor

- converts measurement into electronic signals
- measures:
 - current
 - Vg
 - Temp
 - speed
 - Torque
 - flux

② Interface Circuitry

- conditions signals to microcontroller
- connects sensors / processor & power devices

③ Processor

- processes I/P signals
- O/P signals are amplified

★ Factors Affecting EV propulsion sys:

① Drivetrain Expectation • Acceleration

- max^m speed
- climbing capability

• Braking

• Driving Range

② Vehicle Constraints • Vehicle Vol^m

• Vehicle Weight

③ Energy source

• Payload

• Vehicle Type

• Battery

• Fuel cell

• Ultra-capacitors

• Flywheel

• Hybrid energy sources

★ Single + Multi Motor Drives

Differential: • allows wheels to rotate at different speeds while turning

- outer wheel travels a larger radius than inner wheel.

• Without differential action

↳ wheel slip

↳ Tyre wear

↳ Steering difficulty

↳ poor road holding

Electronic differential:

- EV's can use two or four motors
- Each wheel can have independent motor control.

- wheel speeds are electronically controlled
- provides differential action electronically

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* In-Wheel Drives:

Type 1

(High speed inner Rotor Motor)

Type 2

(Outer-Rotor Motor)

- Requires speed reduction gear
- Usually uses planetary gear
- Gear is placed b/w motor shaft & wheel rim
- Motor can go up to 1000 RPM
- Provides high power density.
- T_a can be completely reversed
- Outer motor acts as wheel rim
- Motor speed = wheel speed
- No gears required.

* In-Wheel Motor Trade-off

INNER-ROTOR MOTOR

Adv: ↳ small size
↳ lower weight
↳ lower cost

OUTER-ROTOR MOTOR

Adv: ↳ low speed → no additional gears required

Disadv: ↳ requires additional planetary gears

↳ larger size
↳ higher weight
↳ higher cost.

* Requirement of EV Motors

- frequent stop/ start
- High rate of acceleration/ deceleration
- High Torque at low speed for hill climbing
- low Torque during cruising
- very wide speed range

★ EV Motors Vs Industrial Motors

① MAXIMUM TORQUE

- EV] 4 x rated Torque
- IM] about 2 x rated Torque

② BASE SPEED

- EV] 4 x Base speed for highway
- IM] 2 x ~~rated~~ ^{base speed}

③ POWER DENSITY & EFFICIENCY

- EV motor requires $\left\{ \begin{array}{l} \text{High power density} \\ \text{High } \eta \text{ over wide} \\ \text{speed / torque range} \end{array} \right.$

- IM] generally optimised

around rated operating point.

④ OPERATING ENVIRONMENT

- EV motors operate under
 - ↳ High Temp
 - ↳ Bad weather
 - ↳ Frequent vibration
- IM] installed in fixed location

★ Classification of EV Motor Drives

① Commutator Types (Traditionally DC motors)

Types: # Series excited

Shunt excited

Compound excited

Separately excited

PM excited.

② Commutatorless:

Induction motor # Synchronous Motor

PM Brushless motor

Switched reluctance motor

PM hybrid motor

★ DC / Commutator Motors

Characteristics # Require commutator & brushes

'Brushes' supplying current to the armature.

Disadvantages # Less reliable

not suitable for maintenance-free operation

not suitable for high speed operation.

Winding - excited DC motors have low

specific power density

#

Advantages # Mature Tech

Simple control

Historically imp. in electric propulsion.

* Commutator less Motors

General:

① Induction Motor:

- # widely used commutatorless motor for EV/HV
- # low cost
- # High Reliability
- # Maintenance free
- # Conventional V/f control may not provide required performance
- # Field Oriented Control

Limitations: # Lower efficiency at light loads
Limited constant power operating range

② PM SM:

- # Field winding replaced by PM
- # Eliminates losses
- # operate with sinusoidal or PWM supply
- # Surface-mounted PM's $\rightarrow$ non-salient motor
- # Interior/Buried PM's $\rightarrow$ saliency produces additional reluctance torque
- # High power density & efficiency

* PM-BLDC

Derived by virtually reversing stator & rotor of PM DC motors

despite the name J - fed by rectangular AC current

NO BRUSHES

produces large torque due to rectangular current flux interaction

More space for

armature windings

Better heat conduction

Sensorless control is also possible.

* SRM :